

Math 202 Analysis Exercises

If $f : \mathbb{R} \rightarrow \mathbb{R}$ is a function that can be integrated (such as $e^x, \sin x, 1 - x^2$, etc) we let $\int_S f(x) dx$ denote the integral of $f(x)$ over the set S . For example, $\int_{[0,1]} f(x) dx = \int_0^1 f(x) dx$ and

$$\int_{[0,1/4] \cup [3/4,1]} f(x) dx = \int_0^{1/4} f(x) dx + \int_{3/4}^1 f(x) dx.$$

- 1.** Give an example of a nonzero function $f(x)$ and a closed set S to such that

$$\int_{[0,1]} f(x) dx = \int_{[0,1] \setminus S} f(x) dx$$

where $[0, 1] \setminus S$ denotes the set $[0, 1]$ without S . Include some indication for why your example works!

- 2.** Give a second example of a nonzero function $f(x)$ and an open set $\mathcal{O}$ such that

$$\int_{[0,1]} f(x) dx = \int_{[0,1] \setminus \mathcal{O}} f(x) dx.$$

Include some indication for why your example works!

- 3.** Take an educated guess of the value of $\int_{[0,1] \cap \mathbb{Q}} x^2 dx$ where $[0, 1] \cap \mathbb{Q}$ is the set of real numbers between 0 and 1 that are not rational (not expressible as a/b for positive integers a, b with $a < b$).

- 4.** Take an educated guess of the value of $\int_C x^2 dx$ where C is the Cantor set.

- 5.** Which of the following numbers are in the Cantor set and why?: $\frac{1}{3}, \frac{1}{4}, \frac{2}{9}, \frac{3}{4}, \frac{1}{\pi}$.